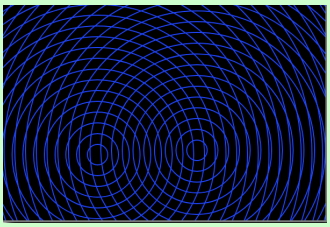
Review: Interference can be constructive or destructive

1. In one dimension, assuming the two sources are coherent

$$\Delta x = m\lambda \quad \text{Constructive Int.}$$
$$\Delta x = \left(m+\frac{1}{2}\right)\lambda \quad \text{Destructive Int.}$$

2. In two dimensions, assuming the two sources are coherent

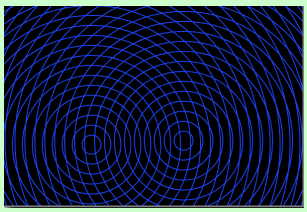
$$\Delta r = m\lambda$$
$$\Delta r = \left(m+\frac{1}{2}\right)\lambda$$



Ex. Two circular waves

In the demo:

1. Offset sources by 2 λ
2. Find places where constructive int happens
3. Connect these to find **Antinodal lines**
4. Find Places were destructive interference happens
5. Connect these to find nodal lines
6. Predict what will happen if the sources are separated by 3 λ
7. Predict what will happen if the wavelength increases

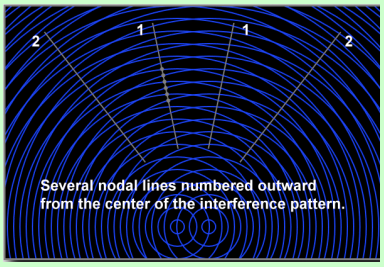


Source Demo

Two source interference:

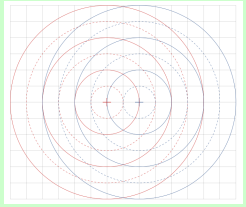
Places where $\Delta r = m\lambda$ Are on the **anti-nodal lines**

$\Delta r = \left(m+\frac{1}{2}\right)\lambda$ Are on the **nodal lines**



Ex. Compasses! The grid spacing is 1cm x 1cm.

Two coherent sources have a wavelength of 2 cm each.
If the sources are separated by 2 cm..



A. Draw solid crests with your compasses for each source
B. Draw dashed troughs with your compasses for each source
C. Identify the nodal and anti-nodal lines
D. Label the lines where one source is $\Delta r = 0, 0.5\lambda, 1\lambda, 1.5\lambda,$ and 2λ closer

Small angle approximation: A result from calculus, called **Taylor Series** allows us to approximate sin x, cos x, and tan x, if the angle Is small (usually less than 10 degrees)

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \approx x \text{ for small angle}$$

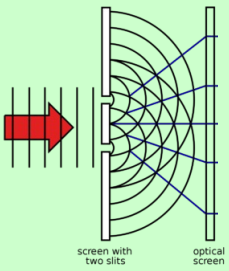
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \approx 1 \text{ for small angle}$$

$$\tan x = \frac{\sin x}{\cos x} = \frac{x - \frac{x^3}{3!} + \cdots}{1 - \frac{x^2}{2!} + \cdots} \approx 1 \text{ for small angle}$$

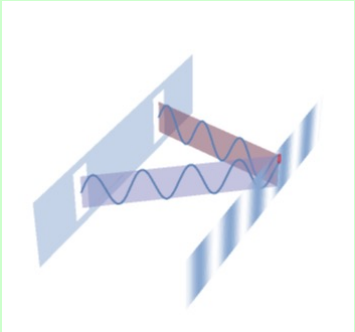
Need to use radian values!!

Ex. Approximate Sin (8°), Tan (4°), Cos (3°)

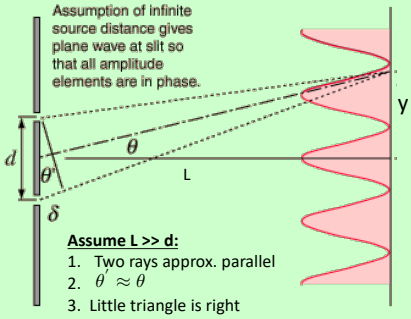
A plane wave is incident on two slits separated by D, what will the intensity pattern on the screen look like?



Mathematica Demo



What intensity pattern forms on the screen?



Assume L >> d:
1. Two rays approx. parallel
2. $\theta' \approx \theta$
3. Little triangle is right

At any point on the screen:

$$\Delta r = \delta = d \sin\theta$$

Maxima: $\boxed{d \sin\theta = m\lambda}$

Minima: $\boxed{d \sin\theta = \left(m + \frac{1}{2}\right)\lambda}$

Use trig to find y:

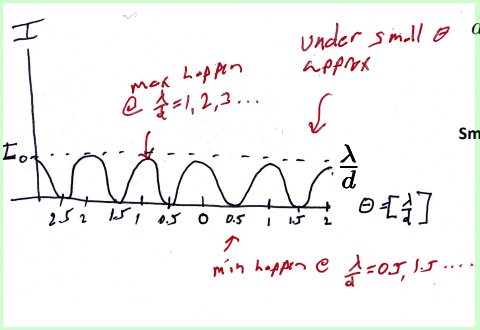
$$\tan\theta = \frac{y}{L}$$

If we know that the angle is small:

$$y = L\theta$$

Maxima happen at $\theta = m\left(\frac{\lambda}{d}\right)$

Maxima happen at $\boxed{y = m\left(\frac{\lambda}{d}\right)L}$



Real Equations for Maxima:

$$d \sin\theta = m\lambda$$
$$y = L \tan\theta$$

Small angle approx. for Maxima:

$$\theta = m\left(\frac{\lambda}{d}\right)$$
$$y = m\left(\frac{\lambda}{d}\right)L$$

Think in terms of $\frac{\lambda}{d}$

Ex. 400 nm light shines on a double slit screen where the two slits are separated by 4 microns. The screen is 20 cm away.

A. Is the small angle approx. valid?

B. Find $\frac{\lambda}{d}$

C. Graph the Intensity pattern vs. angle

D. Graph the Intensity pattern vs position on the screen, y

What is the mathematical formula that calculates the intensity pattern, I?

Recall: $I = \frac{E^2}{2c\mu_0} \implies I = CA^2$

Recall: $A = 2a\cos\left(\frac{\Delta\phi}{2}\right)$

Recall: $\Delta\phi = k\Delta x + \Delta\phi_0$

$$\Delta\phi = \frac{2\pi\Delta r}{\lambda} = \frac{2\pi d \sin\theta}{\lambda}$$

Using the small angle approx.: $\sin\theta = \tan\theta = \frac{y}{L}$

Then: $\frac{\Delta\phi}{2} = \frac{1}{2}\left(\frac{2\pi d(\frac{y}{L})}{\lambda}\right) = \frac{\pi d y}{\lambda L}$

Define: $\boxed{\alpha = \frac{\pi d}{\lambda L}}$ Which implies, $\boxed{\frac{\Delta\phi}{2} = \alpha y}$

$$I = CA^2 = C\left(2a\cos\left(\frac{\Delta\phi}{2}\right)\right)^2 = \boxed{I_0 \cos^2(\alpha y)}$$

Ex. A plane wave with a wavelength of 620 nm shines a double slit, where the slits are spaced by 8 microns. The screen is 10 cm away.

A. Is the small angle approx. valid?
B. At what angle is the m = 3 maxima?
C. How far along the screen is the second minima?
D. Write out the Intensity as a function of position on the screen
E. Show that the first min happens about 4 mm from the central max ?

